Profit Loss Assignment

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1) CP MP SP

$100 \rightarrow 120 \xrightarrow{-\frac{1}{10}} 12\cancel{0} \times \frac{9}{\cancel{10}} = 84$

$120 \xrightarrow{\frac{1}{5}} \overset{24}{\cancel{120}} \times \frac{4}{\cancel{5}} = 96$

$12 = 4.8$

$1 = \frac{\overset{4}{\cancel{4.8}}}{\cancel{12} \times 10}$

$= 0.4$

$100 \times 0.4 = \underline{40}$ (Am) ⓓ

2)

$\underline{OP}$ $\underline{CP}$ SP

$\boxed{100} \rightarrow 90 \longrightarrow 9\cancel{0} \times \frac{12}{1\cancel{0}} = \boxed{108}$

$\frac{8}{100} \times 100$

$= \underline{8\%}$ ⓓ $\underline{(Am)}$

③

26×20
$= 520$

$x \times 30$
$30x$

$39(26 + x)$

$(520 + 30x) \times \frac{\overset{3}{\cancel{150}}}{\underset{2}{\cancel{100}}} = 39(26 + x)$

$\Rightarrow 1560 + 90x = 2(1014 + 39x)$
$= 2028 + 78x$

$\begin{array}{r} 39 \\ \times 26 \\ \hline 1014 \end{array}$

$\begin{array}{r} 90x \\ 78 \\ \hline 12x \end{array}$

$\Rightarrow 12x = \begin{array}{r} 2028 \\ 1560 \\ \hline 0468 \end{array}$

$\boxed{x = 39}$ ⓑ $\underline{(Am)}$

④

$CP = x \quad SP = 1620$
$\downarrow$
$21x$

↳ Profit $= 1620 - 21x$

$1620 - 21x = 6x$

$\Rightarrow \overset{\cancel{3}}{\cancel{27}}x = \overset{\cancel{180}\ 60}{\cancel{1620}}$

$\boxed{x = 60}$ ⓓ

⑤ Ayush → Rahul

$12000 \longrightarrow 120\cancel{00} \times \frac{80}{\cancel{100}}$

$= 9600 \longrightarrow 96\cancel{00} \times \frac{120}{\cancel{100}}$

$= 11520$ ⓒ (Am)

⑥ discount $= 25 + 24 - \frac{\overset{6}{\cancel{24}} \times 25}{\underset{4}{\cancel{100}}}$

$= \underline{43\%}$ ⓑ

⑦

$x \times 25$ 30×30

$30(x + 30)$

$(25x + 900) \times \frac{11}{10} = 30(x + 30)$

$\Rightarrow 275x + 9900 = 300x + 9000$

$\Rightarrow 25x = 900$

$\boxed{x = 36 \text{ kg}}$ ⓒ

$25)900(36$
$\underline{75}$
150

⑧ $33\frac{1}{3}\% = \frac{1}{3}$ $16\frac{2}{3}\% = \frac{1}{6}$

CP MP SP

$\boxed{4500}$ $\overset{1500}{\cancel{4500}} \times \frac{4}{\cancel{3}}$

$= 6000 \times \frac{5}{6} = \boxed{5000}$

Profit $= \underline{500}$ (Am) ⓒ

⑨ Discount $= 20 + 25 - 5$

$= 40\%$

$MP \times \frac{6}{10} = \overset{3000}{\cancel{18000}}$

$\therefore MP = \underline{30000}$ ⓐ (Am)

⑩

CP. $\underline{SP}$

$100 \longrightarrow 110$

$95 \longrightarrow \overset{19}{\cancel{95}} \times \frac{\overset{6}{\cancel{120}}}{\underset{2}{\cancel{100}}} = 114$

$4 = 80$
$1 = 20$

$\therefore$ Cost Price $= 20 \times 100$
$= \underline{2000}$ (Am) ⓒ

Set - 02

① $\underline{CP}$

$15 \rightarrow ₹1$
$1 \rightarrow \frac{1}{15}$

$\underline{SP}$
↳ $\frac{1}{12}$

$\therefore \frac{CP}{SP} = \frac{\frac{1}{15}}{\frac{1}{12}} = \frac{12}{15} = \frac{4}{5}$

$\therefore$ gain $= \frac{1}{\cancel{4}} \times \cancel{100} = \underline{25\%}$ (Am)

②

Discount $= \frac{\text{free}}{\text{Total}} \times 100$ ✓

$= \frac{1}{\cancel{5}} \times \overset{20}{\cancel{100}}$

$= 20\%$

$\therefore$ $\underline{\text{No profit.}}$

③ CP
↳ ₹1/g

CP = 900
SP = 1000

$\therefore$ Profit $= \frac{\cancel{100}}{\cancel{900}} \times 100$

$= 11.11\%$ (Am)

④ $CP_1 + CP_2 = 1008$

↳ $SP_1 = SP_2$

$\Rightarrow CP_1 \times \frac{\overset{4}{\cancel{8}}}{\underset{5}{\cancel{10}}} = CP_2 \times \frac{\overset{18}{\cancel{72}}\ \cancel{144}}{\cancel{100}\ \underset{10}{\cancel{50}}}$

$\Rightarrow \frac{CP_1}{CP_2} = \frac{18}{10} = \frac{9}{5}$

$\therefore CP_1 = \overset{72}{\cancel{1008}} \times \frac{9}{\cancel{14}}$

$= \underline{648}$ (Am)

$14)1008(72$
$\underline{98}$
28
28

⑤ $CP_1 + CP_2 = 5000$

↳ No Profit, No loss

$\therefore$ Total CP = Total SP = 3000

$\Rightarrow SP_1 + SP_2 = 3000$

$\frac{122CP_1}{100} + \frac{92CP_2}{100} = CP_1 + CP_2$

$\Rightarrow \frac{22CP_1}{\cancel{100}} = \frac{8CP_2}{\cancel{100}}$

$\frac{CP_1}{CP_2} = \frac{8}{22} = \frac{4}{11}$

$CP_2 = \overset{200}{\cancel{3000}} \times \frac{11}{\cancel{15}}$

$= \underline{2200}$ (Am)

⑥ CP MP SP

$\boxed{300}$

$3\cancel{00} \times \frac{88}{\cancel{100}} = \boxed{264}$

$CP \times \frac{11\cancel{0}}{10\cancel{0}} = \overset{24}{\cancel{264}}$

$\Rightarrow CP = \boxed{240}$

$\therefore$ Profit $= \frac{\cancel{60}}{\underset{4}{\cancel{240}}} \times \overset{25}{\cancel{100}}$

$= \underline{25\%}$ (Am)

⑦ SP: Shirt + T-shirt $= \boxed{2940}$

$CP \times \frac{\overset{21}{\cancel{105}}}{\underset{20}{\cancel{100}}} = \overset{140}{\cancel{2940}}$

$\therefore CP = \underline{2800}$ $(CP_S + CP_T)$

$21)2940(14$
$\underline{21}$
84

$SP_S + SP_T = 2940$

$CP_S \times \frac{120}{100} + CP_T \times \frac{85}{100} = 2940$

$(CP_S + CP_T) \times \frac{105}{100}$

$\Rightarrow 120CP_S + 85CP_T = 105CP_S + 105CP_T$

$\Rightarrow \overset{3}{\cancel{15}}CP_S = \overset{4}{\cancel{20}}CP_2$

$\frac{CP_S}{CP_T} = \frac{4}{3}$

$\therefore CP_S = \overset{400}{\cancel{2800}} \times \frac{4}{\cancel{7}}$

$= \underline{1600}$ (Am)

8)

CP SP = 700

$\therefore$ Profit $= 700 - CP$ | Profit $= 650 - CP$

$14\frac{2}{7}\% = \frac{1}{7}$

$(700 - CP) = (650 - CP) \times \frac{8}{7}$

$\Rightarrow 4900 - 7CP = 5200 - 8CP$

$\Rightarrow CP = \underline{300}$

$\therefore SP = 3\cancel{00} \times \frac{125}{\cancel{100}}$

$\boxed{SP = 375}$ (Am) ⓓ